

NAME

dispcor -- Dispersion correct and resample spectra

USAGE

dispcor input output [records]

PARAMETERS**input**

List of input spectra or root names to be dispersion corrected. These may be echelle or non-echelle spectra, the task will determine which from the database dispersion functions. When using the record number extension format, record number extensions will be appended to each root name in the list.

output

List of dispersion corrected output spectra or root names. When using the record number extension format, record number extensions will be appended to each root name in the list. The output extension will be the same as the input extension. If "no" output list is specified then the output spectrum will replace the input spectrum after dispersion correction.

records(imred.irsand imred.iidsonly)

List of records or ranges of records to be appended to the input and output root names when using record number extension format. The syntax of this list is comma separated record numbers or ranges of record numbers. A range consists of two numbers separated by a used if no record number extensions are desired. This is a positional query parameter only if the record format is specified.

linearize= yes

Interpolate the spectra to a linear dispersion sampling? If yes, the spectra will be interpolated to a linear or log linear sampling using the linear dispersion parameters specified by other parameters. If no, the nonlinear dispersion function(s) from the dispersion function database are assigned to the input image world coordinate system and the spectral data are not interpolated.

database ="database"

Database containing dispersion solutions created by **identify** or **ecidentify**. If the spectra have been previous dispersion corrected this parameter is ignored unless a new reference spectra are defined.

table = ""

Wavelength coordinate table or reference image. Elements in this optional table or reference image override the wavelength coordinates given below for specified apertures. See the DISCUSSION for additional information.

w1 = INDEF,w2 = INDEF,dw = INDEF,nw = INDEF

The starting wavelength, ending wavelength, wavelength interval per pixel, and the number of pixels in the output spectra. Any combination of these parameters may be used to restrict the wavelength coordinates of the output spectra. If two or more have the value INDEF then suitable defaults based on the number of input pixels and the wavelength range of the reference dispersion solutions are used. These defaults may either come from all spectra, all spectra of the same aperture, or individually for each spectrum depending on the values of the *global* and *samedisp* parameters. Note that these parameters are specified in linear units even if a logarithmic wavelength scale is selected. The conversion between linear and logarithmic intervals between pixels is given below. These values may be overridden for specified apertures by a wavelength table or reference image. Otherwise these values apply to all apertures.

log = no

Transform to linear logarithmic wavelength coordinates? Linear logarithmic wavelength coordinates have wavelength intervals which are constant in the logarithm (base 10) of the wavelength. Note that if conserving flux this will change the flux units to flux per log lambda interval. Note that if the input spectra are in log sampling then *log=no* will resample back to linear sampling and *log=yes* will resample keeping the output spectra in log sampling.

flux = yes

Conserve the total flux during interpolation rather than the flux density? If "no", the output spectrum is average of the input spectrum across each output wavelength coordinate. This conserves flux density. If "yes" the input spectrum is integrated over the extent of each output pixel. This conserves the total flux. Note that in this case units of the flux will change; for example rebinning to logarithmic wavelengths will produce flux per log lambda. For flux calibrated data you most likely would not want to conserve flux.

blank= 0.

Output value corresponding to points outside the range of the input data. In other words, the out of bounds value. This only has an effect when linearizing and the output spectral coordinates extend beyond the input spectral range.

samedisp= no

Use the same dispersion parameters for all apertures? If yes then all apertures in a single image will have the same dispersion parameters. If the *global* parameter is all selected then all spectra in all images will have the same dispersion parameters. This parameter would not normally be used with echelle spectra where each order has a different wavelength coverage.

global = no

Apply global wavelength defaults? Defaults for the INDEF wavelength coordinate parameters are determined if two or less of the wavelength parameters are specified. The defaults are based on the number of pixels and the wavelengths of the first and last pixel as given by the dispersion solution. If this parameter is "no" this is done independently for each input spectrum. If this parameter is "yes" then the maximum number of pixels and the minimum and maximum wavelengths of all the input spectra or those of the same aperture are used to provide defaults for the spectra. The parameter *samedisp* determines whether the global coordinates are over all spectra or only those with the same aperture number. The global option is used to have all the dispersion corrected spectra have the same wavelength coordinates without actually specifying the wavelength parameters.

ignoreaps= no

If a reference dispersion solution is not found for an aperture use the first reference dispersion solution and ignore the aperture number? If not ignoring the apertures all spectra must have a aperture for the dispersion solution and the task aborts if this is not the case. Ignoring the apertures avoids this abort and instead the first dispersion solution is used. Note this parameter does not mean ignore matches between reference and spectrum aperture numbers but only ignore the aperture number if no matching reference is found.

Also if a reference table or image is given and *ignoreaps=yes* then the default dispersion parameters for any aperture not defined by the table or image will be that of the first defined aperture. This can still be overridden by giving explicit values for *w1*, *w2*, *dw* and *nw*.

confirm= no

Confirm the wavelength parameters for each spectrum? If yes the wavelength parameters will be printed and the user will be asked whether to accept them. If the parameters are not acceptable the

user will be queried for new values. The confirmation and parameter changes are repeated until an acceptable set of parameters is obtained. When the *global* parameter is *yes* changes to the wavelength parameters will remain in effect until changed again.

listonly = no

List the dispersion coordinates only? If set then the dispersion coordinates are listed but the spectra are not dispersion corrected. This may be used to determine what the default wavelengths would be based on the dispersion solutions.

verbose = yes

Print the dispersion function and coordinate assignments?

logfile = ""

Log file for recording the dispersion correction operations. If no file name is given then no log information is recorded.

DESCRIPTION

The dispersion coordinate systems of the input spectra are set or changed in the output spectra. The output spectra may be the same as the input spectra if no output spectra are specified or the output name is the same as the input name. The input and output spectra are specified by image templates or lists. In the **irs/iids** packages the input and output spectra are specified as root names and the record numbers are specified by the *record* parameter. The records are given as a set of comma separate single numbers or ranges of hyphen separated numbers. If no records are specified then the input and output images are assumed to be full names.

The dispersion coordinate system is defined either in the image header or by dispersion functions in the specified database. To use reference spectra dispersion functions they must first be assigned to the image with **identify** (**reidentify**), **ecidentify** (**ecreidentify**), **refspectra**, or **hedit**. These tasks define the image header keywords REFSPEC1, REFSPEC2, REFSHFT1, and REFSHFT2. The test which determines whether to use the current dispersion coordinate system or reference spectra dispersion solutions is the presence of the REFSPEC1 keyword. Since it is an error to apply a dispersion function to data which have already been dispersion corrected the any dispersion function keywords are deleted after use and a record of them entered in sequential image header keywords beginning with DCLOG.

Dispersion functions are specified by one or both of the reference spectrum image header keywords REFSPEC1 and REFSPEC2 containing the name of calibration spectra with dispersion function solutions (either echelle dispersion functions from **ecidentify** or non-echelle dispersion functions from **identify**) in the database. There must be a dispersion function for each aperture in the input spectrum unless the *ignoreaps* flag is set. If the flag is not set the task will abort if a aperture is not found while if it is set spectra without a matching aperture in the reference dispersion solutions will use the first dispersion solution. Note that aperture number matching is done in both cases and the *ignoreaps* parameter only applies to non-matching spectra. The common situation for using the *ignoreaps* option is there is a single reference dispersion solution which is to be applied to a number of spectra with different aperture numbers; hence effectively ignoring the reference spectrum aperture number.

If two reference spectra are specified the names may be followed by a weighting factor (assumed to be 1 if missing). The wavelength of a pixel is then the weighted average of the wavelengths of the two dispersion functions. The task **refspectra** provides a number of ways to assign reference spectra. Note, however, that these assignments may be made directly using the task **hedit** or with some other task or script if none of the methods are suitable. Also note that **identify** and **reidentify** add the REFSPEC1 keyword referring to the image itself when a database entry is written.

In addition to the one or two reference dispersion functions for each input aperture there may also be image header keywords REFSHFT1 and REFSHFT2 specifying reference spectra whose dispersion function zero

point shifts (the "shift" parameter in the database files) are to be applied to the reference dispersion functions. The shifts from REFSHFT1 will be applied to the dispersion functions from REFSPEC1 and similarly for the second dispersion functions. The reference shifts need not be present for every aperture in a multispectrum image. By default the mean shift from all the reference apertures having a zero point shift is applied to all the reference dispersion functions. If the REFSHFT keyword has the modifier word "nearest" following the spectrum name then the shift from the nearest aperture in spatial position (from the aperture extraction limits in the original 2D spectrum as recorded in the 6th and 7th fields of the APNUM keywords) is used for a particular input aperture. If the modifier word is "interp" then the nearest two apertures are used to interpolate a zero point shift spatially.

The purpose of the reference shift keywords is to apply a point correction to the reference dispersion functions determined from separate arc calibration observations using a same time as object observations. For example, consider multifiber observations in which one or more fibers are assigned to arc lamps at the same time the other fibers are used to observe various objects. The basic dispersion reference, the REFSPEC keywords, will come from arc observations taken through all the fibers. The arc fibers used during an object observation are then calibrated against their corresponding fibers in the arc calibration observations to determine a zero point shift. The REFSHFT keywords will contain the name of the object spectrum itself and the shifts from the simultaneous arc fibers will be interpolated spatially to the nonarc object fibers and applied to the dispersion functions from the arc calibrations for those fibers.

The reference shift keywords are currently added with **hedit** and zero point shifts computed with **identify/reidentify**. The complexities of this have been hidden in the multifiber **imred** instrument reduction packages. The reference shift correction feature was added primarily for use in those reduction packages.

If the *linearize* parameter is no the dispersion functions, weights, and shifts are transferred from the database to the world coordinate system keywords in the image header. Except for printing processing information that is all that is done to the spectra.

If the *linearize* parameter is yes the spectra are interpolated to a linear wavelength scale and the dispersion coordinate system in the header is set appropriately. A linear wavelength coordinate system is defined by a starting wavelength, an ending wavelength, a wavelength interval per pixel, and the number of pixels. These four parameters actually overspecify the coordinate system and only three of these values are needed to define it. The output coordinate system is specified by giving a these parameters using the parameters *w1*, *w2*, *dw*, and *nw*.

When the *log* option is used these parameters are still specified and computed in non-log units but the effective interval per pixel is

$$\begin{aligned} dw_log &= (\log_{10}(w2) - \log_{10}(w1)) / (nw - 1) \\ dw_log &= (\log_{10}(w1+dw*(nw-1)) - \log_{10}(w1)) / (nw - 1) \end{aligned}$$

In other words, the logarithmic interval divides the starting and ending wavelength into the required number of pixels in log step. To avoid confusion in this case it is best to specify the starting and ending wavelengths (in non-log units) and the number of pixels.

Note that if *log*=yes the input spectra in either linear or log sampling will be resampled to produces an output spectrum in log sampling. Similarly, if *log*=no the input spectra will be resampled to linear sampling. This means that log sampled input spectra will be resampled to linear sampling.

Default values for any parameters which are not specified, by using the value INDEF, are supplied based on the wavelengths of the first and last pixel as given by the dispersion function and the number of pixels in the input image. The defaults may either be determined separately for each spectrum (*global* = no), from all spectra with the same aperture (*global* = yes and *samedisp* = no), or from all the spectra (*global* = yes and *samedisp* = yes). As indicated, the parameter *samedisp* determines whether defaults are determined independently for each aperture or set the same for all apertures.

Another way to specify the wavelengths when there are many apertures is use a wavelength table or reference image. If an spectrum image name is specified with the *table* parameter then the dispersion parameters for each apertures are set to be the same as the reference spectrum. Alternatively, a text file table consisting of lines containing an aperture number, the starting wavelength, the ending wavelength, the wavelength interval per pixel, and the number of output pixels may be specified. Any of these values may be specified as INDEF (though usually the aperture number is not). One way to view the wavelength table/reference spectrum is that an entry in the wavelength table/reference spectrum overrides the values of the parameters *w1*, *w2*, *dw*, and *nw*, which normally apply to all apertures, for the specified aperture. The wavelength table is used to specify explicit independent values for apertures. The global mechanism can supply independent values for the INDEF parameters when the *samedisp* parameter is no.

If one wishes to verify and possibly change the defaults assigned, either globally or individually, the *confirm* flag may be set. The user is asked whether to accept these values. By responding with no the user is given the chance to change each parameter value. Then the new parameters are printed and the user is again asked to confirm the parameters. This is repeated until the desired parameters are set. When the defaults are not global the changed parameters will not be used for the next spectrum. When the global option is used any changes made are retained (either for all apertures or independently for each aperture) until changed again.

When adjusting the wavelengths the user should specify which parameter is free to change by entering INDEF. If none of the parameters are specified as INDEF then those values which were not changed, i.e. by accepting the current value, are the first to be changed.

Once the wavelength scale has been defined the input spectrum is interpolated for each output pixel. Output wavelengths outside the range of the input spectrum are set to the value given by the *blank* parameter value. The default interpolation function is a 5th order polynomial. The choice of interpolation type is made with the package parameter "interp". It may be set to "nearest", "linear", "spline3", "poly5", or "sinc". Remember that this applies to all tasks which might need to interpolate spectra in the **onedspec** and associated packages. For a types see **onedspec**.

When it is desired to conserve total flux, particularly when the dispersion is significantly reduced, the parameter *flux* is set to yes and the output pixel value is obtained by integrating the interpolation function across the wavelength limits of the output pixel. If it is set to no then the flux density is conserved by averaging across the output pixel limits.

The input spectrum name, reference spectra, and the wavelength parameters will be printed on the standard output if the *verbose* parameter is set and printed to a log file if one is specified with the *logfile* parameter. If one wishes to only check what wavelengths will be determined for the defaults without actually dispersion correcting the spectra the *listonly* flag may be set.

Other tasks which may be used to change the dispersion coordinate system are **scopy**, **specshift**, and **sapertures**.

EXAMPLES

In the examples when the task is used in the IRS and IIDS packages, shown with the "ir>" prompt the spectra have a record number extension image name format and the records parameter must be specified. In the other case shown with the "on>" prompt the records parameter is not used.

1. Dispersion correct spectra so that they have the same number of pixels and the wavelengths limits are set by the reference spectra.

```
ir> dispcor spec dcspec 9,10,447-448
dcspec.0009: ap = 0, w1 = 5078.84, w2 = 6550.54, dw = 1.797, nw = 820
dcspec.0010: ap = 1, w1 = 5078.71, w2 = 6552.81, dw = 1.800, nw = 820
dcspec.0447: ap = 0, w1 = 5082.57, w2 = 6551.45, dw = 1.794, nw = 820
dcspec.0448: ap = 1, w1 = 5082.03, w2 = 6553.66, dw = 1.797, nw = 820
```

```

on> dispcor allspec.ms dcallspec.ms
dcallspec.ms: ap = 1, w1 = 5078.84, w2 = 6550.54, dw = 1.797, nw = 820
dcallspec.ms: ap = 2, w1 = 5078.71, w2 = 6552.81, dw = 1.800, nw = 820
dcallspec.ms: ap = 3, w1 = 5082.57, w2 = 6551.45, dw = 1.794, nw = 820
dcallspec.ms: ap = 4, w1 = 5082.03, w2 = 6553.66, dw = 1.797, nw = 820

```

2. Confirm and change assignments.

```

on> dispcor spec* %spec%new%* confirm+
new009: ap = 0, w1 = 5078.84, w2 = 6550.54, dw = 1.797, nw = 820
Change wavelength coordinate assignments? (yes):
Starting wavelength (5078.8421234): 5070
Ending wavelength (6550.535123):
Wavelength interval per pixel (1.79693812):
Number of output pixels (820): INDEF
new009: ap = 0, w1 = 5070., w2 = 6550.53, dw = 1.795, nw = 826
Change wavelength coordinate assignments? (yes): no
new010: ap = 1, w1 = 5078.71, w2 = 6552.81, dw = 1.800, nw = 820
Change wavelength coordinate assignments? (no): yes
Starting wavelength (5078.7071234): 5100
Ending wavelength (6550.805123): 6500
Wavelength interval per pixel (1.79987512): INDEF
Number of output pixels (820): INDEF
new010: ap = 1, w1 = 5100., w2 = 6500., dw = 1.797, nw = 780
Change wavelength coordinate assignments? (yes): no
new447: ap = 0, w1 = 5082.57, w2 = 6551.45, dw = 1.793, nw = 820
Change wavelength coordinate assignments? (yes): no
new448: ap = 1, w1 = 5082.03, w2 = 6553.66, dw = 1.797, nw = 820
Change wavelength coordinate assignments? (no):

```

3. Confirm global assignments and do dispersion correction in place. record format.

```

ir> dispcor irs "" 9,10,447,448 confirm+ global+ samedisp+
irs.0009: ap = 0, w1 = 5078.71, w2 = 6553.66, dw = 1.801, nw = 820
Change wavelength coordinate assignments? (yes):
Starting wavelength (5078.7071234): 5100
Ending wavelength (6553.664123): 6500
Wavelength interval per pixel (1.80092412):
Number of output pixels (820):
irs.0009: ap = 0, w1 = 5100., w2 = 6500., dw = 1.799, nw = 779
Change wavelength coordinate assignments? (yes): no
irs.0010: ap = 1, w1 = 5100., w2 = 6500., dw = 1.799, nw = 779
Change wavelength coordinate assignments? (no):
irs.0447: ap = 0, w1 = 5100., w2 = 6500., dw = 1.799, nw = 779
Change wavelength coordinate assignments? (no):
irs.0448: ap = 1, w1 = 5100., w2 = 6500., dw = 1.799, nw = 779
Change wavelength coordinate assignments? (no):

```

4. Make a nonlinear dispersion correction in place.

```

on> dispcor spec* "" linearize=no verbose- logfile=logfile

```

5. Apply a single dispersion solution to a set of record number format images.

```

ir> dispcor nite101 dcnite101 "1-10" ignore+ confirm-

```

REVISIONS

DISPCORV2.12.3

Added the blank parameter value.

DISPCORV2.11.3

Long slit and data cubes can be used with this task to either resample using the existing WCS or to use a single dispersion function from IDENTIFY. It uses the first one found.

DISPCORV2.10.3

Provision was added for IDENTIFY dispersion solutions consisting of only a shift (as produced by the 'g' key in IDENTIFY or the refit=no flag in REIDENTIFY) to be applied to previously LINEARIZED spectra. Thus it is possible to use IDENTIFY/REIDENTIFY to automatically compute a zero point shift based on 1 or more lines and then shift all the spectra to that zero point.

DISPCOR will now allow multiple uses of IDENTIFY dispersion solutions in a simple way with but with continuing protection against accidental multiple uses of the same dispersion solutions. When a first dispersion corrected using one or more reference spectra keywords the dispersion flag is set and the reference spectra keywords are moved to DCLOGn keywords. If DISPCOR is called again without setting new reference spectra keywords then the spectra are resampled (rebinned) using the current coordinate system. If new reference spectra are set then DISPCOR will apply these new dispersion functions. Thus the user now explicitly enables multiple dispersion functions by adding reference spectra keywords and DISPCOR eliminates accidental multiple uses of the same dispersion function by renaming the reference spectra. The renamed keywords also provide a

The flux conservation option now computes an average across the output pixel rather than interpolating to the middle of the output pixel when *flux* is no. This preserves the flux density and includes all the data; i.e. a coarse resampling will not eliminate features which don't fall at the output pixel coordinates.

Some additional log and verbose output was added to better inform the user about what is done.

Better error information is now printed if a is not found.

DISPCORV2.10

This is a new version with many differences. It replaces the previous three tasks **dispcor**, **ecdispcor** and **msdispcor**. It applies both one dimensional and echelle dispersion functions. The new parameter *linearize* selects whether to interpolate the spectra to a linear dispersion (the only option available previously) or to assign a nonlinear dispersion function to the image without any interpolation. The interpolation function parameter has been eliminated and the package parameter *interp* is used to select the interpolation function. The new interpolation type "sinc" may be used but care should be exercised. The new task supports applying a secondary zero point shift spectrum to a master dispersion function and a spatial interpolation of the shifts when calibration spectra are taken at the same time on a different region of the same 2D image. The optional wavelength table may now also be an image to match dispersion parameters. The *apertures* and *rebin* parameters have been eliminated. If an input spectrum has been previously dispersion corrected it will be resampled as desired. Verbose and log file parameters have been added to log the dispersion operations as desired. The record format syntax is available in the **irs/iids** packages.

SEE ALSO

package, refspectra, scopy, specshift, sapertures